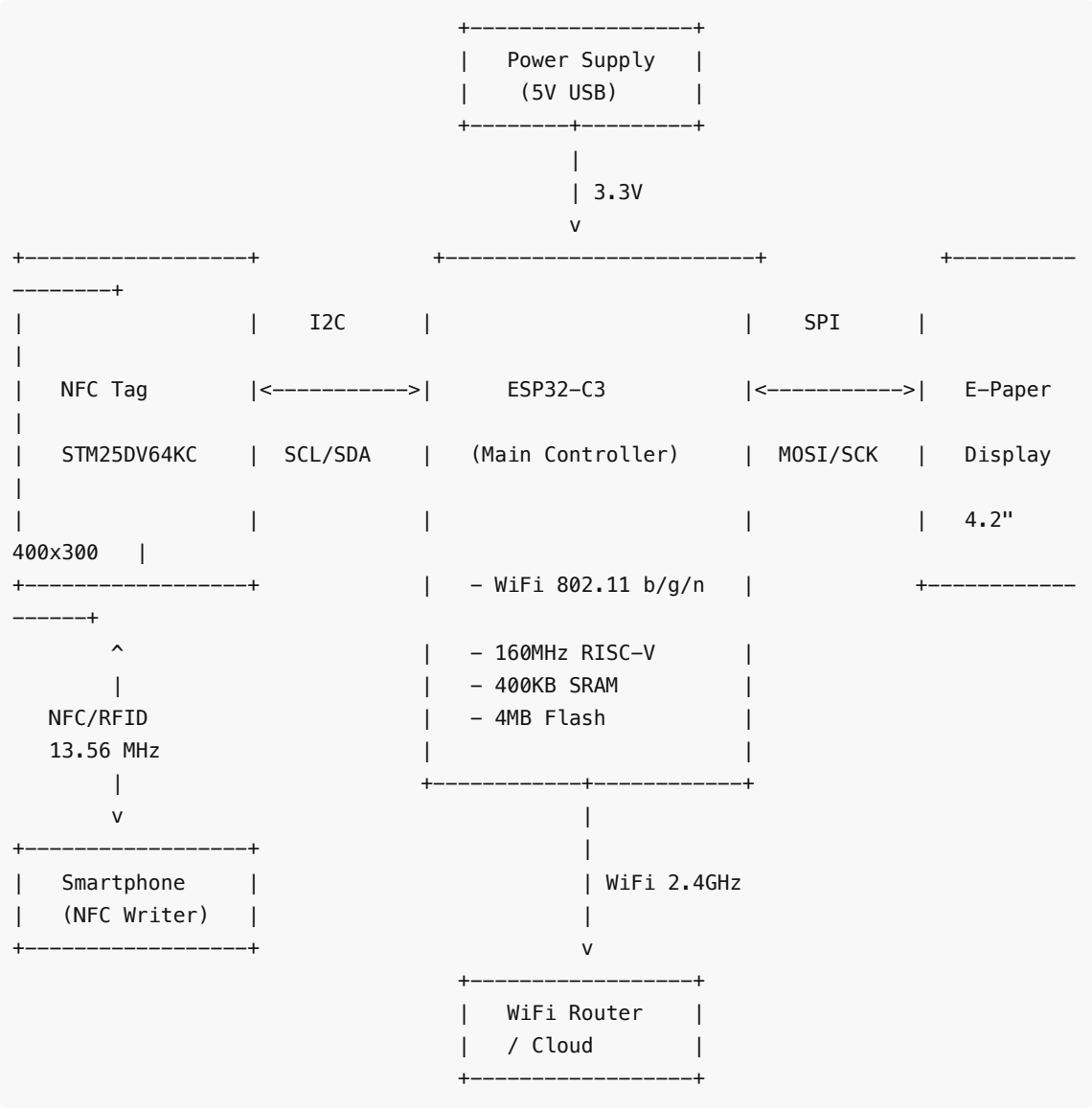


WiFi Display - Hardware Block Diagram

Product Overview

Product Name: WiFi Display **Model:** WD-01 **Description:** WiFi-enabled e-paper display with NFC provisioning

System Block Diagram



Detailed Pin Connections

ESP32-C3 Module (Pre-certified)

GPIO	Function	Connected To	Interface
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GPIO2	Output	Status LED	Digital
GPIO3	Output	NFC VCC Control	Digital
GPIO4	Output	E-Paper RST	Digital
GPIO5	Output	E-Paper DC	Digital
GPIO6	I2C SCL	NFC Tag	I2C
GPIO7	I2C SDA	NFC Tag	I2C
GPIO8	SPI SCK	E-Paper CLK	SPI
GPIO9	Input	NFC GPO	Digital
GPIO10	SPI MOSI	E-Paper DIN	SPI
GPIO20	Input	E-Paper BUSY	Digital
GPIO21	Output	E-Paper CS	SPI

Component Specifications

1. Main Controller - ESP32-C3

Parameter	Value
Manufacturer	Espressif
Module	ESP32-C3-WROOM-02 (or similar)
CPU	32-bit RISC-V, 160MHz
WiFi	802.11 b/g/n, 2.4GHz
Bluetooth	BLE 5.0 (available, not used)
Flash	4MB
SRAM	400KB
Operating Voltage	3.0V - 3.6V
Operating Temp	-40C to +85C
FCC ID	(Check module certification)
CE	(Check module certification)

2. E-Paper Display - 4.2" EPD

Parameter	Value
Display Type	E-Paper / E-Ink

Resolution	400 x 300 pixels
Size	4.2 inches diagonal
Colors	Black/White
Interface	SPI (4-wire)
Refresh Time	~2-4 seconds
Operating Voltage	3.3V
Driver IC	SSD1683 or similar

3. NFC Tag - STM25DV64KC

Parameter	Value
Manufacturer	STMicroelectronics
Type	Dynamic NFC/RFID EEPROM
Memory	64Kbit (8KB)
RF Frequency	13.56 MHz
Interface	I2C (up to 1MHz)
NFC Standard	ISO 15693, NFC Forum Type 5
Operating Voltage	1.8V - 5.5V
Operating Temp	-40C to +85C

4. Power Supply

Parameter	Value
Input Voltage	5V DC (USB)
Regulated Output	3.3V
Current (Active WiFi)	~120-350mA
Current (Deep Sleep)	<10uA

RF Information

WiFi (ESP32-C3)

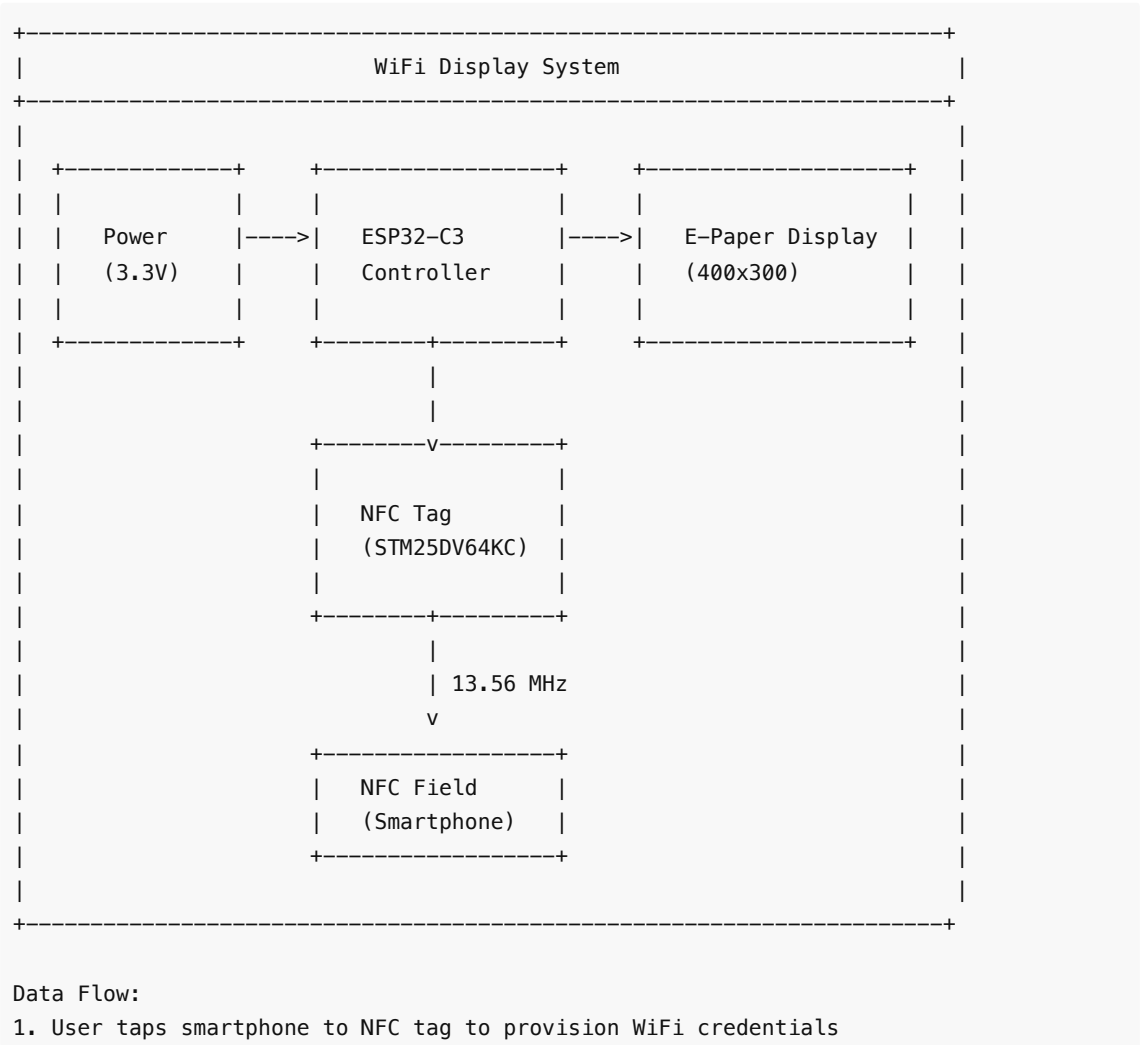
Parameter	Value
Frequency Band	2.4 GHz (2400 - 2483.5 MHz)
Modulation	OFDM, DSSS

TX Power	Up to +21 dBm
Standards	IEEE 802.11 b/g/n
Antenna	PCB Antenna / Chip Antenna
Pre-certified	Yes (Espressif module)

NFC (STM25DV64KC)

Parameter	Value
Frequency	13.56 MHz
Protocol	ISO 15693
Range	<5cm (passive)
Note	Passive tag, no intentional radiator

Functional Block Diagram



2. ESP32-C3 reads credentials from NFC tag via I2C
3. ESP32-C3 connects to WiFi network
4. ESP32-C3 receives display content via MQTT over WiFi
5. ESP32-C3 renders content to E-Paper display via SPI

Interfaces Summary

Interface	Speed	Devices
SPI	20 MHz	E-Paper Display
I2C	100 kHz (standard)	NFC Tag (STM25DV64KC)
WiFi	802.11n (up to 150Mbps)	Cloud/Router
NFC	13.56 MHz	Smartphone (external)

Compliance Notes

1. **WiFi Module:** Uses pre-certified ESP32-C3 module from Espressif

- Antenna design follows Espressif reference design
- No modifications to RF section

2. **NFC Tag:** Passive device (STM25DV64KC)

- No intentional RF emissions
- Powered by external NFC field

3. **EMC Considerations:**

- SPI and I2C traces kept short
 - Proper decoupling capacitors on power rails
 - Ground plane under RF section
-

Contact Information

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